

Chapter 5 Explanation

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PDF

Explain every topic in this chapter. Don't skip anything.

Alright — no shortcuts, no skipping. I'll break **every topic of Chapter 5: *Jacobians – Velocities and Static Forces*** into **simple, exam-ready understanding**, step by step.

I'll follow the exact structure of your chapter so you don't miss anything.

(Everything is based on your uploaded file)

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◆ 5.1 INTRODUCTION

What this chapter is about:

- Earlier: you studied **position (where robot is)**
- Now: we study:
 - **Velocity (how fast it moves)**
 - **Forces (how strong it pushes)**

Key Idea:

☞ Both velocity and force are connected using a matrix called:

★ Jacobian (J)

- It links:
 - Joint motion → End-effector motion
 - Forces → Torques

☞ This is the **heart of robotics motion + control**

◆ 5.2 NOTATION (Important Foundation)

1. Differentiation of position

Velocity = derivative of position:

$$v = \frac{dQ}{dt}$$

👉 But in robotics:

- Everything depends on **frames (A, B, C)**

So velocity depends on:

- Which frame you observe from
 - Which frame you differentiate in
-

2. Frame Notation (CRITICAL)

Example from page 2-3:

- ${}^A v_B$: velocity of B **expressed in A**
- ${}^C v_{ORG}$: velocity of origin

👉 If frames change → velocity representation changes

3. Angular Velocity (Ω)

- Linear velocity → motion in straight line
- Angular velocity → rotation

👉 Represented as vector:

$$\Omega = [\omega_x, \omega_y, \omega_z]$$

- Direction → axis of rotation
 - Magnitude → speed of rotation
-

◆ 5.3 LINEAR & ROTATIONAL VELOCITY

1. Linear Velocity

From page 4:

$$v_Q = v_{origin} + R \cdot v_{relative}$$

👉 Meaning:

- Velocity of a point =
 - velocity of frame origin
 - relative motion inside frame
-

2. Rotational Velocity

If frame rotates:

$$\frac{dQ}{dt} = \Omega \times Q$$

👉 This is **VERY IMPORTANT**

Velocity due to rotation = **cross product**

3. Combined Motion (Most Important)

From page 6:

$$v_Q = v_{origin} + \Omega \times r$$

👉 Interpretation:

- Object moving + rotating
 - Final velocity = translation + rotation
-

◆ 5.4 MORE ON ANGULAR VELOCITY

1. Property of Rotation Matrix

$$R^T R = I$$

Differentiating:

👉 Leads to:

$$\dot{R}R^T = S$$

Where:

- S = skew-symmetric matrix
-

2. Skew-Symmetric Matrix

$$S = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}$$

👉 This matrix represents **angular velocity**

3. Cross Product Form

$$SP = \Omega \times P$$

👉 So:

- Matrix form = cross product form
-

4. Physical Meaning

From page 9:

👉 Angular velocity:

- Has **axis + speed**
 - Represents **instantaneous rotation**
-

◆ 5.5 MOTION OF ROBOT LINKS

- Each link has:
 - Linear velocity v_i
 - Angular velocity ω_i

👉 Always measured in base frame (0)

◆ 5.6 VELOCITY PROPAGATION (SUPER IMPORTANT)

👉 Robot = chain of links

We compute velocity **link by link**

Key Equations:

Angular velocity:

$$\omega_{i+1} = \omega_i + \text{contribution from joint}$$

Linear velocity:

$$v_{i+1} = v_i + \omega_i \times P$$

👉 Meaning:

- Velocity flows from base → end-effector

Types of joints:

1. Revolute (rotation)

- Adds angular velocity

2. Prismatic (sliding)

- Adds linear velocity

◆ 5.7 JACOBIANS (CORE TOPIC)

Definition:

Jacobian = matrix of partial derivatives

$$J = \frac{\partial Y}{\partial X}$$

In Robotics:

$$v = J(\theta)\dot{\theta}$$

Where:

- $\dot{\theta}$ = joint velocity
- v = end-effector velocity

👉 This is THE main equation

Structure:

$$v = \begin{bmatrix} \text{linear} \\ \text{angular} \end{bmatrix}$$

👉 Jacobian maps:

- Joint space $\rightarrow$ Cartesian space

Key Insight:

- Jacobian changes with time
 - Because robot configuration changes
-

◆ 5.8 SINGULARITIES (VERY IMPORTANT FOR EXAM)

What is singularity?

👉 When Jacobian is NOT invertible

$$\det(J) = 0$$

Why dangerous?

- Cannot control robot in some direction
 - Infinite joint velocity needed
-

Types:

1. Workspace Boundary

- Arm fully stretched

2. Interior Singularities

- Links aligned
-

Example (page 18):

- When $\theta_2 = 0$ or 180°
👉 Robot loses DOF
-

◆ 5.9 STATIC FORCES

Now we switch from velocity → forces

Key Idea:

Forces travel **backward** (end → base)

Force balance:

$$f_i = f_{i+1}$$

Torque balance:

$$n_i = n_{i+1} + P \times f$$

Joint torque:

$$\tau = J^T F$$

👉 VERY IMPORTANT

- Force → torque via **Jacobian transpose**
-

◆ 5.10 JACOBIAN IN FORCE DOMAIN

Virtual Work Principle:

$$F \cdot dx = \tau \cdot d\theta$$

👉 Leads to:

$$\tau = J^T F$$

Insight:

- Jacobian:
 - velocity → forward
 - force → backward
-

◆ 5.11 CARTESIAN TRANSFORMATION

6×6 representation

Velocity:

$$V = \begin{bmatrix} v \\ \omega \end{bmatrix}$$

Force:

$$F = \begin{bmatrix} f \\ n \end{bmatrix}$$

Transformation:

$$V_B = TV_A$$

$$F_A = T^T F_B$$

👉 Velocity and force transform differently

🔥 FINAL BIG PICTURE (DON'T MISS THIS)

Jacobian connects EVERYTHING:

Input	Output
Joint velocity	End velocity
End force	Joint torque

💥 What You Must Remember for Exam

If you forget everything, remember this:

1.

$$v = J\dot{\theta}$$

2.

$$\tau = J^T F$$

3.

Singularity:

$$\det(J) = 0$$

⚠️ Brutal Reality Check

If you:

- Don't understand cross product → you're weak in kinematics

- Don't understand Jacobian → you **cannot solve robotics problems**

Want Next Step?

I can:

- ♦ Turn this into **PDF revision sheet**
- ♦ Solve **numerical problems step-by-step**
- ♦ Give **past paper questions + answers**
- ♦ Make **1-day exam crash plan**

Just tell me.

Sources

- ♦ Solve numerical problems step-by-step

Good — this is where most students break. If you can't solve numericals, theory is useless.
I'll walk you through **step-by-step solved problems** (exactly like exam style). No skipping.

PROBLEM 1: Velocity using Jacobian (2-Link Planar Robot)

Given:

- 2-link robot
- $L_1 = 1, L_2 = 1$
- $\theta_1 = 30^\circ, \theta_2 = 45^\circ$
- $\dot{\theta}_1 = 1 \text{ rad/s}, \dot{\theta}_2 = 2 \text{ rad/s}$

 Find end-effector velocity

Step 1: Position Equations

$$x = L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2)$$

$$y = L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2)$$

Step 2: Jacobian Matrix

$$J = \begin{bmatrix} -L_1 \sin \theta_1 - L_2 \sin(\theta_1 + \theta_2) & -L_2 \sin(\theta_1 + \theta_2) \\ L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2) & L_2 \cos(\theta_1 + \theta_2) \end{bmatrix}$$

✓ Step 3: Plug Values

Convert:

- $\theta_1 = 30^\circ, \theta_2 = 45^\circ$
- $\theta_1 + \theta_2 = 75^\circ$

Use:

- $\sin 30 = 0.5, \cos 30 = 0.866$
 - $\sin 75 \approx 0.966, \cos 75 \approx 0.259$
-

Compute J:

$$J = \begin{bmatrix} -(1 \times 0.5 + 1 \times 0.966) & -0.966 \\ (0.866 + 0.259) & 0.259 \end{bmatrix}$$

$$J = \begin{bmatrix} -1.466 & -0.966 \\ 1.125 & 0.259 \end{bmatrix}$$

✓ Step 4: Multiply

$$v = J\dot{\theta}$$

$$\dot{\theta} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Final:

$$v_x = (-1.466)(1) + (-0.966)(2) = -3.398$$

$$v_y = (1.125)(1) + (0.259)(2) = 1.643$$

✓  Answer:

$$v = \begin{bmatrix} -3.398 \\ 1.643 \end{bmatrix}$$

PROBLEM 2: Joint Torque from Force

Given:

- Same robot
- Force at end-effector:

$$F = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$$

👉 Find joint torques

✓ **Step 1: Use formula**

$$\tau = J^T F$$

✓ **Step 2: Transpose J**

$$J^T = \begin{bmatrix} -1.466 & 1.125 \\ -0.966 & 0.259 \end{bmatrix}$$

✓ **Step 3: Multiply**

First torque:

$$\begin{aligned} \tau_1 &= (-1.466)(10) + (1.125)(5) \\ &= -14.66 + 5.625 = -9.035 \end{aligned}$$

Second torque:

$$\begin{aligned} \tau_2 &= (-0.966)(10) + (0.259)(5) \\ &= -9.66 + 1.295 = -8.365 \end{aligned}$$

✓ 🎯 **Answer:**

$$\tau = \begin{bmatrix} -9.035 \\ -8.365 \end{bmatrix}$$

🔥 **PROBLEM 3: Singularity Check**

👉 When does robot lose control?

✓ Step 1: Determinant of Jacobian

$$\det(J) = L_1 L_2 \sin \theta_2$$

✓ Step 2: Condition

$$\det(J) = 0 \Rightarrow \sin \theta_2 = 0$$

✓ 🎯 Final Answer:

$$\theta_2 = 0^\circ \text{ or } 180^\circ$$

👉 Robot is:

- Fully stretched OR folded
 - 👉 Loses 1 DOF
 - 👉 Cannot move in some direction
-

🔥 PROBLEM 4: Velocity from Angular Motion

Given:

- $\Omega = [0, 0, 2]$
 - $r = [1, 0, 0]$
-

✓ Step:

$$v = \Omega \times r$$

Cross product:

$$v = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$$

✓ 🎯 Answer:

$$v = [0, 2, 0]$$

💀 Brutal Reality Check

If you:

- Struggle with trig → you will lose marks
- Don't remember Jacobian form → you're dead in exam
- Don't practice multiplication → too slow under pressure

🚀 Next Level (Your Choice)

I can push you harder:

♦ Option 1:

Give you **PAST PAPER level numericals**

♦ Option 2:

Give you **timed test (10 mins / question)**

♦ Option 3:

Make **PDF with 20 solved + 20 practice problems**

Just say:

👉 *"give me harder problems"*

Start from example 1 in the document that I provided you and explain all examples.

Perfect — now we go **exactly from your document examples**, no skipping. I'll explain each example **step-by-step in simple words**, and also tell you **what examiner expects**.

🔥 ✅ EXAMPLE 5.1 (Page 3) – Frame Velocity Transformation

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🧠 Scenario:

- Train moving at **100 mph**
- Car moving at **30 mph**
- Both move in **x-direction**
- Frames:

- $\{U\} \rightarrow$ universe (ground)
 - $\{B\} \rightarrow$ train
 - $\{C\} \rightarrow$ car
-

? What is asked:

Find velocities in different frames

✓ Step 1: Velocity of car origin in universe

$$\frac{d}{dt}({}^U P_{CORG}) = V_{CORG} = 30\hat{x}$$

👉 Simple:

- Car moves at 30 mph in ground frame
-

✓ Step 2: Velocity of train relative to car

$${}^C V_{TBORG} = R \cdot (100\hat{x}) = R^{-1}100\hat{x}$$

👉 Meaning:

- Transform velocity into another frame using rotation matrix
-

✓ Step 3: Change frame

$${}^C V_{CORG} = R^T V_{CORG}$$

👉 Key idea:

- Velocity **depends on frame**
 - Same motion looks different in different frames
-

🎯 What examiner wants:

- You understand:
 - Velocity is **frame-dependent**
 - Use of **rotation matrix**
-

🔥 ✅ EXAMPLE 5.2 (Page 10)

🧠 Scenario:

Convert Euler angles → angular velocity

✅ Key Result:

$$\Omega = E(\theta)\dot{\theta}$$

👉 Where:

- E = matrix
 - Converts **angle rates** → **angular velocity**
-

🎯 Key Insight:

- Angular velocity ≠ simple derivative of angles
 - Needs transformation matrix
-

🔥 ✅ EXAMPLE 5.3 (Page 12–14) – Two-Link Robot Velocity

(VERY IMPORTANT — HIGH EXAM PROBABILITY)

🧠 Scenario:

- 2-link manipulator
 - Compute:
 - Linear velocity
 - Angular velocity
-

✅ Step 1: Define transformations

From page 13:

$${}^0T_1, {}^1T_2, {}^2T_3$$

👉 These describe:

- How each link is connected

✓ Step 2: Angular velocity propagation

$$\omega_1 = [0, 0, \dot{\theta}_1]$$

$$\omega_2 = \omega_1 + [0, 0, \dot{\theta}_2]$$

✓ Step 3: Linear velocity

Using:

$$v = \omega \times r$$

👉 Each link velocity depends on:

- Previous link
- Rotation effect

✓ Step 4: Final result

From page 14:

$$v_3 = \begin{bmatrix} l_1 \dot{\theta}_1 + l_2 (\dot{\theta}_1 + \dot{\theta}_2) \\ 0 \end{bmatrix}$$

🎯 What examiner expects:

- Apply:
 - Velocity propagation
 - Cross product
 - Link-by-link calculation

🔥 ✓ EXAMPLE 5.4 (Page 18) – Singularity Analysis

🧠 Scenario:

- Two-link robot
- Find when singularity occurs

✓ Step 1: Determinant

$$\det(J) = l_1 l_2 \sin \theta_2$$

✓ Step 2: Set zero

$$\sin \theta_2 = 0$$

🎯 Answer:

$$\theta_2 = 0 \text{ or } 180^\circ$$

💡 Meaning:

- Robot becomes straight
 - Loses motion in one direction
-

🔥 ✓ EXAMPLE 5.5 (Page 18–19)

🧠 Scenario:

- Robot moving in **x-direction**
 - Near singularity
-

✓ Key Result:

$$\dot{\theta}_1, \dot{\theta}_2 \rightarrow \infty$$

💀 Meaning:

- As robot approaches singularity:
 - Required joint speed → **infinite**
-

🎯 Examiner trap:

👉 "Explain why singularity is dangerous"

Answer:

- Infinite velocity needed
 - System becomes unstable
-

🔥 ✅ **EXAMPLE 5.6 (Page 19)**

🧠 **Scenario:**

- PUMA 560 robot
 - Different singularities
-

✅ **Types:**

1. Wrist singularity
 2. Arm stretched
 3. Internal alignment
-

🎯 **Key Idea:**

- Multiple types of singularities exist in real robots
-

🔥 ✅ **EXAMPLE 5.7 (Page 21–22) – Static Forces**

🧠 **Scenario:**

- Robot applies force at end-effector
-

✅ **Step 1: Force propagation**

$$f_i = f_{i+1}$$

✅ **Step 2: Torque**

$$n_i = n_{i+1} + P \times f$$

✓ Step 3: Final result

$$\tau = \begin{bmatrix} l_1 f_y + l_2 f_y \\ l_2 f_y \end{bmatrix}$$

🎯 Key Concept:

$$\tau = J^T F$$

👉 THIS is the core result

🔥 ✓ EXAMPLE 5.8 (Page 25) – Force Transformation

🧠 Scenario:

- Force sensor at wrist
 - Convert forces between frames
-

✓ Key Equation:

$$F_A = T^T F_B$$

🎯 Insight:

- Velocity → transform using T
 - Force → transform using T^T
-

💀 FINAL REALITY CHECK

If you don't deeply understand:

- Example 5.3 → you don't understand velocity propagation
- Example 5.7 → you don't understand force domain
- Example 5.4 → you don't understand singularities

👉 Then you are **not ready for exam**

🚀 What Next?

I can push you harder (recommended):

- ♦ **1. Convert all examples into numerical practice problems**
- ♦ **2. Give past paper solved questions**
- ♦ **3. Make 1-page cheat sheet (exam killer)**

Just say:

👉 *"test me now"*

Sources